FE Graded Quiz M4

Question 1

Given the following MA(1) model:

$$X_t = 30 + W_t + 0.4W_{t-1},$$

what is $\rho(1)$?

- 0.255
- 0.768
- 0.534
- 0.344

Question 2

There are 3 blue balls, 2 red balls, and 5 green balls in a box. You draw two balls one after the other, and you do not put the ball back after each draw. What is the probability that the second ball is green?

- 1/3
- 1/5
- 4/5
- 1/2

Question 3

Which of the following statements is **false** regarding AIC and BIC?

- AIC and BIC measure the goodness of fit for a model by balancing between the error of the model and number of parameters.

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- The smaller the values of AIC or BIC, the better the model fits the data.
- AIC and BIC are information criteria to select models.
- AIC puts more emphasis on penalizing the number of parameters.

Question 4

What is the mean of the following AR(1) process: $X_t = 3 + 0.75X_{t-1} + W_t$ where W_t is normal white noise with mean = 0 and variance = 1?

- 12
- 3
- 24
- 4

Question 5

We have $X_t = 0.7X_{t-3} + W_t + 0.5W_{t-1}$. Which of the following expressions is correct for X_t in terms of the backshift operator B and W_t?

- $X_t = \frac{(1-0.5B)}{(1-0.7B^3)}W_t$
- $X_t = \frac{(1+0.5B)}{(1-0.7B^3)}W_t$
- $X_t = \frac{(1-0.5B)}{(1+0.7B^3)}W_t$
- $X_t = \frac{(1+0.5B)}{(1+0.7B^3)}W_t$

Question 6

Which of the following time series is stationary?

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- $X_t = X_{t-1} + W_t$ where W_t is iid normal (0,1)
- $X_t = c + X_{t-1} + W_t$ where c is constant and W_t is iid normal (0,1)
- $X_t = c + b*t$ where b and c are constants
- $X_t = c + W_t + a*W_{t-1}$ where a and c are constants and W_t is iid normal (0,1)

Question 7

Which of the following is the downside of using ARIMA models to analyze a time series?

- It is difficult to select independent variables.
- It is difficult to create a forecast from the model.
- ARIMA models are not based on an economic or financial model.
- It cannot handle a trend component in the data.

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Question 8

What is the requirement for an AR process to be stationary?

- All roots of the characteristic equation need to be greater than 1 in absolute value.
- All roots of the characteristic equation need to be smaller than 1.
- All roots of the characteristic equation need to be greater than 1.
- All roots of the characteristic equation need to be smaller than 1 in absolute value.

Question 9

Which of the following is **false** regarding random walk with a drift?

- [ ] The sign of the drift does not decide the overall trend of a random walk.
- [ ] We can use random walk with a drift to model a time series with a trend.
- [ ] Random Walk with a drift is non-stationary.
- [x] The first difference of random walk with a drift is a random walk.

Question 10

The following table shows the number of different types of cookies in a jar. You took a cookie out. What is the probability of getting a white chocolate cookie when you know the cookie does not have nuts?

	Dark Chocolate	Milk Chocolate	White Chocolate
With Nuts	6	4	2
No Nuts	3	4	1

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- 1/3
- 3/20
- 2/12
- 1/8

QUESTIONS

1 2 3 4
5 6 7 8
9 10 11 12
13 14 15 16

Question 11

Which of the following statements is **false** regarding the differencing method?

- Second difference can remove quadratic trend in time
- Second difference for X_t is equal to X_t - X_{t-2}
- First difference can remove linear trend in time
- It is used to remove trending non-stationary behavior in a time series

Question 12

Which of the following statements regarding the Yule-Walker estimation vs. maximum likelihood estimation is **false**?

- The coefficients of maximum likelihood estimation have lower variance.
- Both methods use every individual data point from the time series for calculation.
- Maximum likelihood estimation is computationally intensive.
- Yule-Walk estimation only applies to an AR model.

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Question 13

Which of the following is equal to $(1-B)^2 X_t$ where B is a backshift operator?

- $2X_t + 2X_{t-1}$
- $X_t + 2X_{t-1} - X_{t-2}$
- $X_t + X_{t-1} + X_{t-2}$
- $X_t - 2X_{t-1} + X_{t-2}$

Question 14

Given P(A) = 0.3, P(B) = 0.2, and P(A|B) = 0.1, what is P(B|A)?

- 0.333
- 0.666
- 0.500
- 0.067

Question 15

Which of the following methods cannot make a time series stationary?

- Transform the data with log or square root
- Differencing
- Standardize the data
- Estimate linear time trend and remove it

Question 16

Given the following ACF plot, which model would you select to model the time series?

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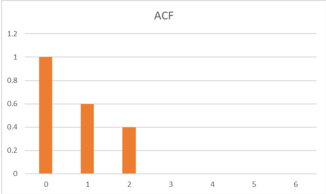

- AR(1)
- MA(2)
- MA(1)
- AR(2)

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